PROJECT REPORT

Smart EV Battery Management System with SOH Prediction

A Course project report submitted in partial fulfilment of requirement for the course

On

EMBEDDED SYSTEMS

By

P.BHAVANI	(2405A41L03)
K.HARINI	(2405A41L04)
A.GEETHANJALI	(2305A41043)
J.ABHINAY	(2405A41L15)
J.ROHITH	(2305A41012)

Under the guidance of

Dr Sandip Bhattacharya

Prof., Department of ECE

SR UNIVERSITY

CERTIFICATE

This is to certify that the course project entitled "Smart EV Battery Management System with SOH Prediction" is the Bonafide work carried out by P.BHAVANI (2405A41L03), K.HARINI (2405A41L04),A.GEETHANJALI (2305A41043), J.ABHINAY (2405A41L15),J.ROHITH (2305A41012in the partial fulfilment of the requirement for the award of course Embedded System during the academic year 2025-2026 under our guidance and Supervision.

Dr. Sandip Bhattacharya

Prof., Department of ECE

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Abstract

This project presents a Smart Electric Vehicle (EV) Battery Management System (BMS) integrated with advanced State of Health (SOH) prediction techniques to improve battery performance, safety, and lifespan. The system continuously monitors critical battery parameters such as temperature, voltage, current, and state of charge (SOC) in real time using embedded sensors.

The collected data is transmitted to a cloud platform using Firebase, enabling remote monitoring, storage, and analysis. Machine learning algorithms are employed to process historical and real-time data for accurate prediction of battery health, remaining useful life (RUL), discharge patterns, and potential failure risks. This predictive capability allows early detection of abnormalities such as overheating, overcharging, and rapid discharge, thereby preventing hazardous conditions.

The system also incorporates intelligent decision-making features that provide maintenance alerts and optimize battery usage through efficient charge–discharge control strategies. A prototype EV model is developed, integrating sensors, a microcontroller-based control unit, and a display interface to demonstrate real-time visualization of battery status and system responses.

Overall, the proposed smart BMS enhances battery reliability, ensures safety, reduces maintenance costs, and supports sustainable energy utilization, making it highly suitable for next-generation electric vehicle applications.

LIST OF FIGURES

1. Introduction

Electric Vehicles (EVs) are increasingly replacing conventional vehicles due to their environmental benefits and energy efficiency. However, the performance, safety, and reliability of EVs depend heavily on the battery system. Lithium-ion batteries, commonly used in EVs, are sensitive to factors such as temperature, overcharging, deep discharge, and irregular current flow.

A Battery Management System (BMS) plays a crucial role in ensuring optimal battery operation by continuously monitoring and controlling battery parameters. Traditional BMS systems mainly focus on protection and monitoring, but lack predictive intelligence.

This project enhances a conventional BMS by integrating:

- **Cloud computing (Firebase)** for remote monitoring and data storage
- **Machine Learning (ML)** for predictive analysis of battery health
- **Real-time visualization system** for user interaction

The proposed system ensures improved safety, efficiency, and intelligent decision-making.

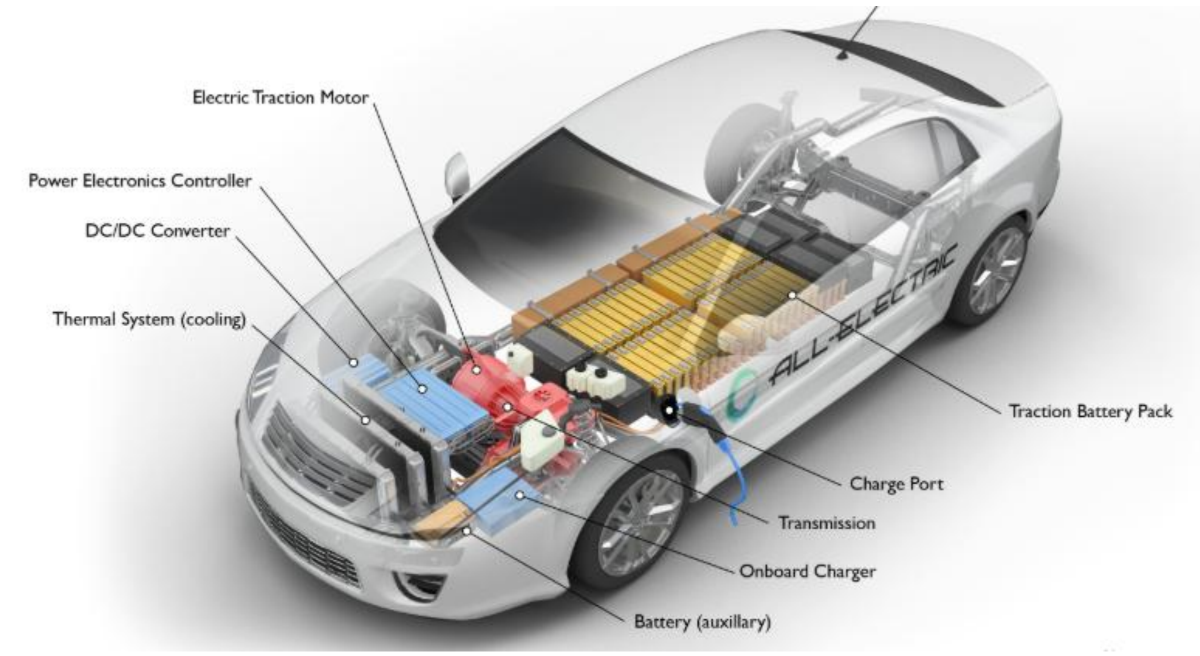


Fig a : Internal Architecture of an Electric Vehicle Showing Battery Pack, Motor, Power Electronics, and Thermal Management System.

2. Objectives

The main objectives of this project are:

- To design a **real-time battery monitoring system** using sensors
- To implement **temperature-based protection mechanisms** to avoid thermal runaway
- To develop a **machine learning model** for predicting:
    - Battery life
    - Remaining charge cycles
    - Driving range
- To provide **early warning alerts** for abnormal conditions such as overheating or over-discharge
- To enable **IoT-based remote monitoring** using Firebase cloud platform
- To create a **user-friendly dashboard** for visualization and analysis.

Fig b: Electric Vehicle Powertrain Layout with Key Components.

3. System Architecture

3.1 Additional Explanation:

The system follows a layered architecture:

3.1.1. Data Acquisition Layer

- Sensors collect physical parameters:
  - Temperature sensor (DHT11/LM35)
  - Voltage sensor
  - Current sensor (ACS712)

3.1.2. Processing Layer

- Microcontroller (Arduino/ESP32):
  - Reads sensor data
  - Processes threshold conditions
  - Sends data to cloud

3.1.3. Communication Layer

- Wi-Fi module (ESP8266/ESP32)
- Sends data to Firebase in real time

3.1.4. Cloud Layer

- Firebase stores:
  - Historical data
  - Real-time updates
- Enables remote access

3.1.5. Application Layer

- Dashboard displays:
  - Battery status

- Alerts
- Predictions

3.1.6. Intelligence Layer

- Machine Learning model:
  - Trained using battery datasets
  - Predicts SOH, RUL, and range

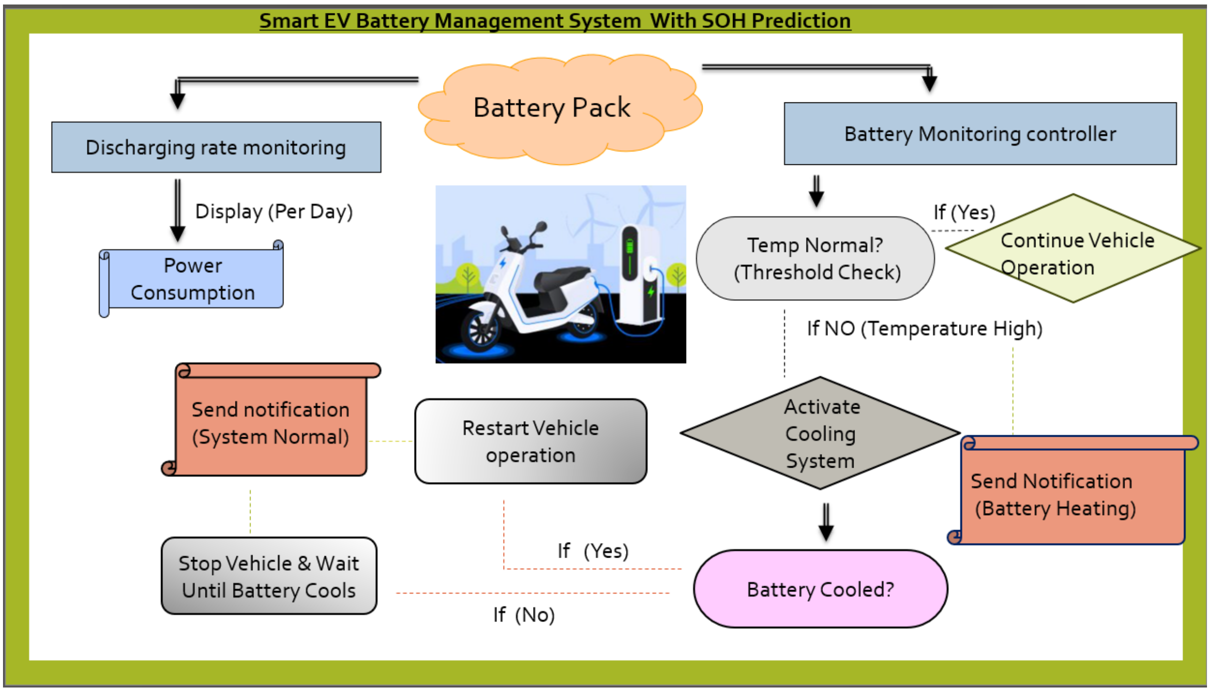


Fig C : System Architecture and Working Flow of Smart EV Battery Management System with SOH Prediction

4. Hardware Implementation

4.1 Additional Components :

- Current sensor (ACS712)
- Voltage divider circuit
- Cooling fan (for overheating control)
- Relay module (to control charging/discharging)

4.2 Working:

- Sensors continuously read battery parameters
- Microcontroller processes values
- LCD shows live status
- Buzzer activates during abnormal conditions

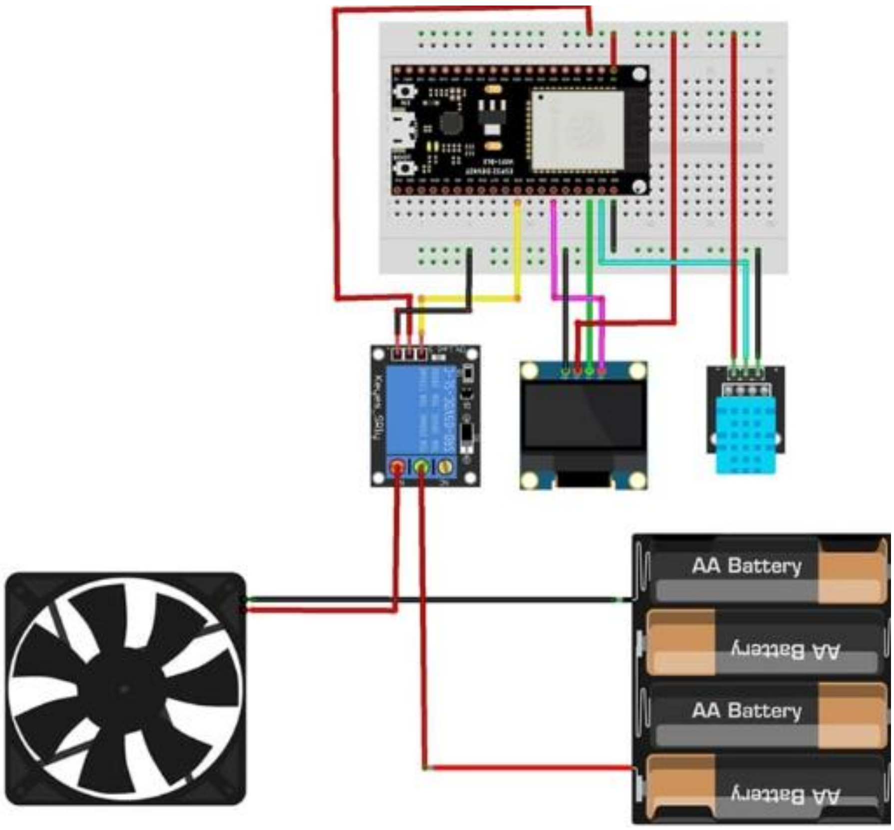


Fig d: Hardware Circuit Diagram of Smart EV Battery Management System Using ESP Microcontroller, Sensors, Relay Module, and Cooling Fan.

5. Software Implementation

5.1 Backend:

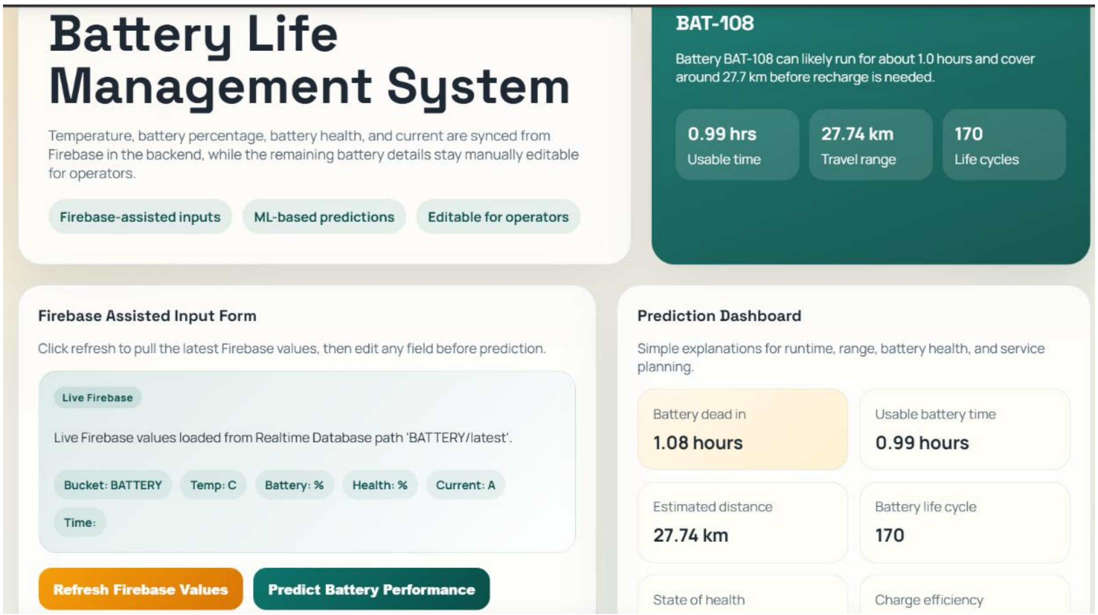
- Firebase Realtime Database:
    - Stores sensor data
    - Syncs data instantly

5.2 Frontend Dashboard:

- Can be built using:
    - Web (HTML/CSS/JS)
    - Python (Streamlit)
    - Android app

5.3 Features:

- Graphical visualization (charts)
- Historical trend analysis
- Alert logs
- Prediction confidence display.



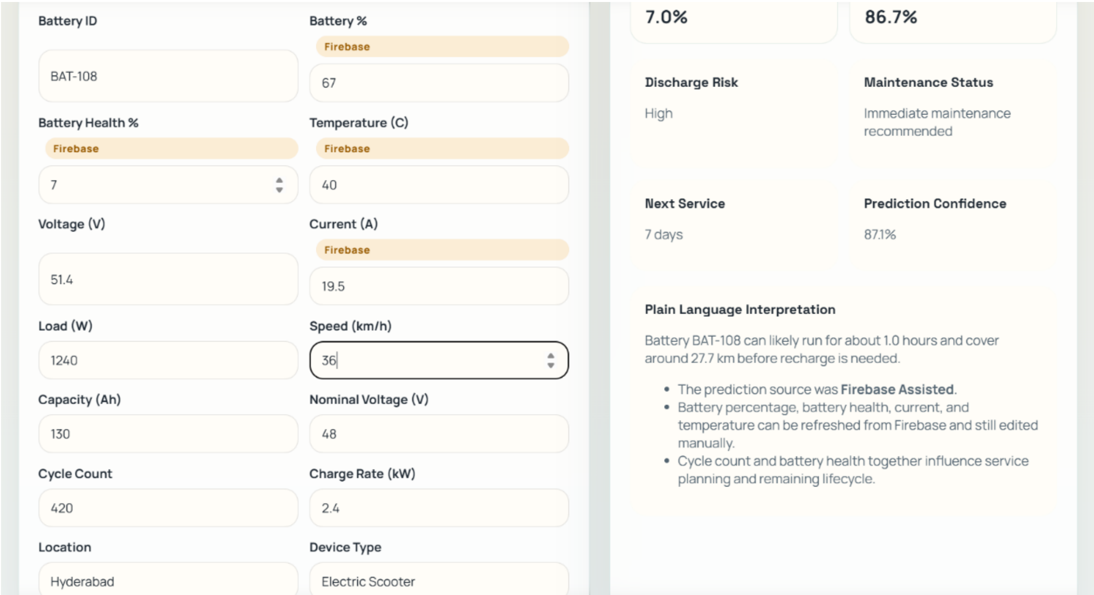


Fig e: Software Dashboard for Smart EV Battery Management System Showing Firebase Data Integration and ML-Based Battery Performance Prediction.

6. Machine Learning Model

6.1 Model Types:

- Linear Regression (for battery life prediction)
- Random Forest (for better accuracy)
- Neural Networks (advanced option)

6.2 Training Process:

1. Data collection (sensor + dataset)
2. Data preprocessing:
    - Normalization
    - Handling missing values
3. Model training
4. Evaluation:
    - Accuracy

- Mean Squared Error (MSE)

6.3 Additional Outputs:

- Failure probability
- Health degradation rate

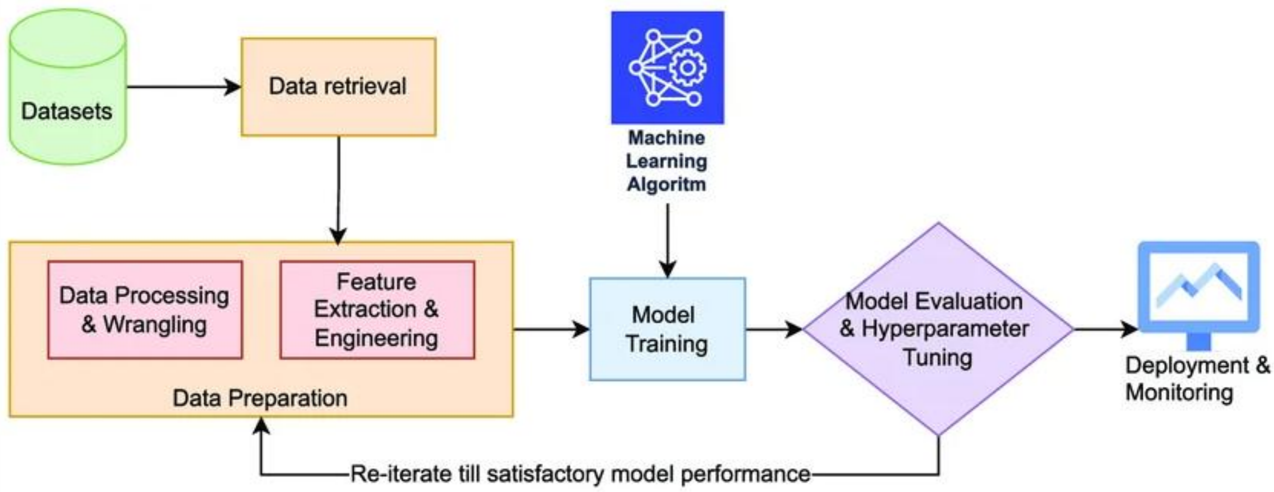


Fig f: Machine Learning Workflow for Battery State of Health (SOH) and Performance Prediction.

7. Working Principle

The system operates in a continuous loop:

1. Sensors measure battery parameters
2. Microcontroller reads and processes data
3. Data is:
    - Displayed locally (LCD)
    - Sent to Firebase cloud
4. Threshold comparison:
    - If temperature exceeds limit → cooling system ON
    - If voltage drops → alert generated

5. ML model processes data to predict:
    - Remaining battery life
    - Driving range
    - Maintenance schedule

Fig g: Prototype Model Demonstrating Real-Time Battery Monitoring and Display in Smart EV Battery Management System.

8. Advantages

- Improves **battery safety** by preventing overheating
- Enables **predictive maintenance**, reducing failures
- Enhances **battery lifespan and efficiency**
- Provides **real-time and remote monitoring**
- Reduces operational costs
- Scalable for real-world EV systems

9. Applications

- Electric cars and buses
- E-bikes and scooters
- Renewable energy storage (solar batteries)
- Uninterruptible Power Supply (UPS)
- Smart grid systems

10. Limitations

- Accuracy depends on quality of ML training data
- Requires stable internet for cloud functionality
- Prototype may not reflect real EV complexity
- Limited battery size in demo model
- Sensor errors may affect predictions

11. Future Enhancements

- Mobile application (Android/iOS)
- Integration with GPS for route-based prediction
- AI-based adaptive charging system
- Fast charging optimization
- Thermal management using advanced cooling
- Integration with real EV battery packs.

12. Conclusion

This project successfully demonstrates an intelligent Battery Management System integrating IoT and Machine Learning. The system not only monitors battery parameters in real time but also predicts future battery performance, ensuring enhanced safety, efficiency, and reliability.

The combination of cloud computing and predictive analytics makes the system highly suitable for next-generation electric vehicles and smart energy systems.

13. References

You can format like this for reports:

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2. Arduino Official Documentation – https://www.arduino.cc
3. Scikit-learn Library – https://scikit-learn.org
4. TensorFlow Documentation – https://www.tensorflow.org
5. Research Papers on EV Battery Management Systems (IEEE, Elsevier)